

2D Long-Range Ising Universality Boundary

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Scientific question

Is the boundary at 7/4 or 2?

Sak scenario

$$\sigma^* = 2 - \eta_{\text{SR}} = \frac{7}{4}$$

At the critical 2D short-range Ising fixed point, the exact spin scaling dimension $x_s = \frac{1}{8}$ gives $\langle s(0)s(r) \rangle \sim r^{-2x_s} = r^{-\frac{1}{4}}$, hence $\eta_{\text{SR}} = \frac{1}{4}$.

The Sak criterion then gives $\sigma^* = 2 - \eta_{\text{SR}} = \frac{7}{4}$.

Geometric scenario

$$\sigma^* = 2$$

In momentum space, the long-range kernel contributes a nonanalytic k^σ term that dominates k^2 for $\sigma < 2$, suggesting that the local short-range term takes over only at $\sigma = 2$.

The task is not to choose the prettiest fit, but to test whether the data exclude either boundary.

Why is the crossover hard to resolve?

1. **Slow drift** Near the boundary, a small correction exponent delays the asymptotic regime. [→ scale](#)
2. **Model degeneracy** Power and near-marginal logarithmic drifts can both look linear over finite sizes. [→ form](#)
3. **Fragile intercept** Similar fit quality can imply different thermodynamic limits. [→ limit](#)

Decision rule

A thermodynamic claim requires both correction-model agreement and fit-window stability.

Model, observables, and calibration

The model is a 2D long-range Ising ferromagnet

Hamiltonian

$$H = - \sum_{i < j} J_{ij} s_i s_j, \quad J_{ij} = \frac{c(\sigma, L)}{r_{ij}^{2+\sigma}}, \quad s_i = \pm 1$$

1. **Geometry** $L \times L$ square torus with the minimum-image distance r_{ij} .

→ lattice

2. **Normalization** $c(\sigma, L)$ is chosen so that $\sum_j J_{ij} = 4$, summing over all other sites j .

→ scale

3. **Range control** The minus sign makes the interaction ferromagnetic; increasing σ suppresses distant bonds and drives the model toward short-range Ising behavior.

→ physics

FK topology gives a dimensionless critical condition

$$R_p \equiv \langle R_2 \rangle - q \langle R_0 \rangle, \quad q = 2$$

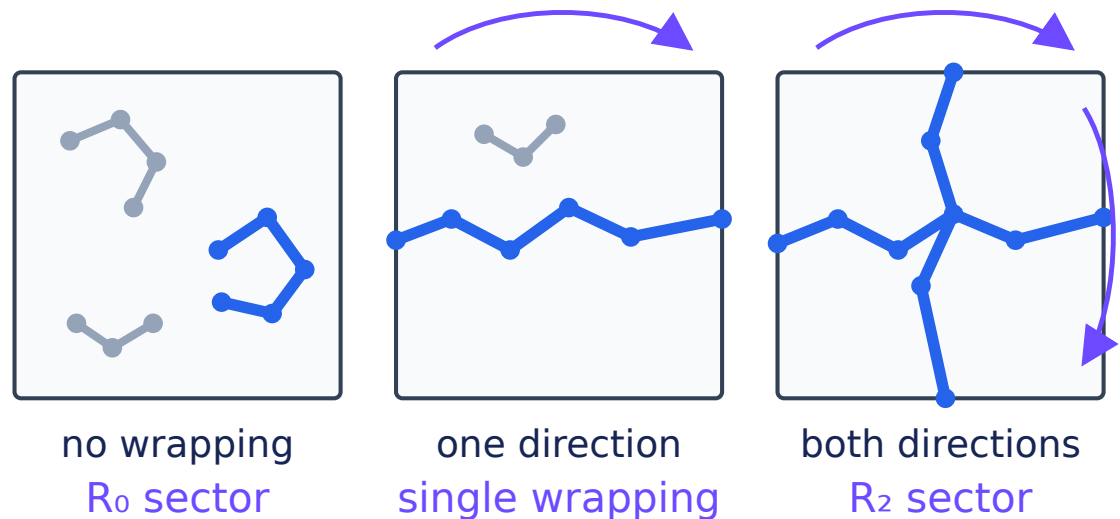
Topological events

R_0 : no FK cluster wraps around the torus.

R_2 : an FK cluster wraps in both periodic directions.

$R_p = 0$ means that the two topological sectors satisfy the random-cluster critical balance $\langle R_2 \rangle = q \langle R_0 \rangle$.

FK-cluster topology on the torus



Q_m records the magnetization-distribution shape

$$Q_m \equiv \frac{\langle M^2 \rangle^2}{\langle M^4 \rangle}$$

Disordered phase

$P(M)$ is approximately a Gaussian centered at zero.

$$\langle M^4 \rangle = 3\langle M^2 \rangle^2$$

$$Q_m \rightarrow \frac{1}{3}$$

Critical point

For fixed geometry and boundary conditions, $P(M)$ has a universal non-Gaussian finite-size shape.

$$Q_m^{\text{SR}} \approx 0.856216$$

Ordered phase

$P(M)$ approaches two narrow peaks at $\pm M_0$.

$$\langle M^4 \rangle \approx \langle M^2 \rangle^2$$

$$Q_m \rightarrow 1$$

Reduced susceptibility

Definition

$$\tilde{\chi} \equiv L^2 \langle m^2 \rangle$$

Motivation Measures the growth of critical magnetic fluctuations and yields η through finite-size scaling.

Disordered phase

Short-range correlations; the magnetization density self-averages.

$$\langle m^2 \rangle \sim L^{-2}$$

$$\tilde{\chi} \sim O(1)$$

Critical point

Scale-free correlations. For 2D short-range Ising, $\eta = \frac{1}{4}$.

$$\tilde{\chi} \sim L^{\frac{7}{4}}$$

Ordered phase

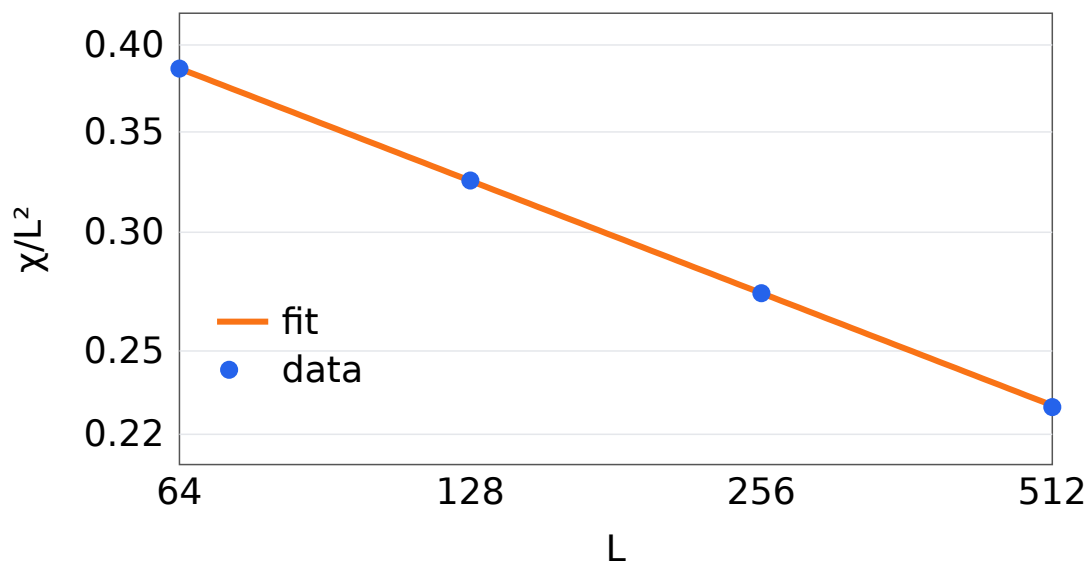
A finite spontaneous magnetization survives as $L \rightarrow \infty$.

$$\langle m^2 \rangle \rightarrow m_0^2$$

$$\tilde{\chi} \sim m_0^2 L^2$$

The strict NN model recovers $\eta=1/4$

Figure 2 · Strict-NN calibration



$\frac{\chi(L)}{L^2}$ follows one power law across all four sizes on logarithmic axes.

$$\frac{\chi(L)}{L^2} \sim AL^{-\eta}$$

$$\eta = 0.24883$$

95% bootstrap interval:

[0.24748, 0.25016]

The exact 2D nearest-neighbor
Ising critical point has $\eta = \frac{1}{4}$.

Three observables provide complementary constraints

FK topology

$$R_p = \langle R_2 \rangle - 2\langle R_0 \rangle$$

Probes system-spanning cluster connectivity and supplies a dimensionless critical condition.

Magnetization shape

$$Q_m = \frac{\langle M^2 \rangle^2}{\langle M^4 \rangle}$$

Compresses the non-Gaussian shape of $P(M)$ into a universal dimensionless ratio.

Reduced susceptibility

$$\tilde{\chi} \equiv L^2 \langle m^2 \rangle$$

Measures the growth of critical magnetic fluctuations and yields η through finite-size scaling.

Geometry, distribution shape, and fluctuation scaling test the same critical interpretation from different directions.

Official baseline

The official $L \leq 512$ grid reproduces drift, not the boundary

σ	published β_c ¹	β_c from R_p crossings	β_c from Q_m crossings
1.75	0.329136	0.329148	0.329434
1.875	0.336985	0.337304	0.337593
2.0	0.344439	0.344857	0.345165
2.5	0.369446	0.370195	0.370435

¹ T. Xiao, Z. Liu, Z. Fan, and Y. Deng, On Sak's criterion for statistical models with long-range interaction, arXiv:2512.04805 (2025).

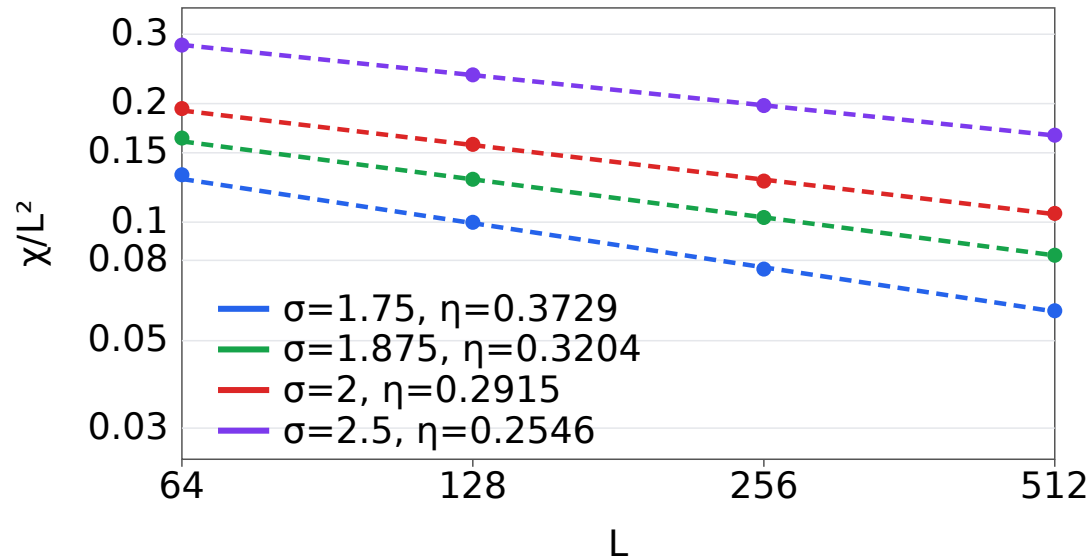
Every crossing lies inside the registered $\beta_c \pm 0.002$ window, reproducing the expected critical-point drift.

The R_p and Q_m estimates differ slightly because their finite-size corrections are different; neither crossing is itself a universality-boundary estimate.

| The baseline validates the phenomenon but cannot identify the correction law.

Effective exponents on the official grid shift toward the short-range value

Figure 1 · Finite-size fits of η on the official $L \leq 512$ grid



Across four sampled σ values, the fitted exponent shifts toward the short-range benchmark $\frac{1}{4}$.

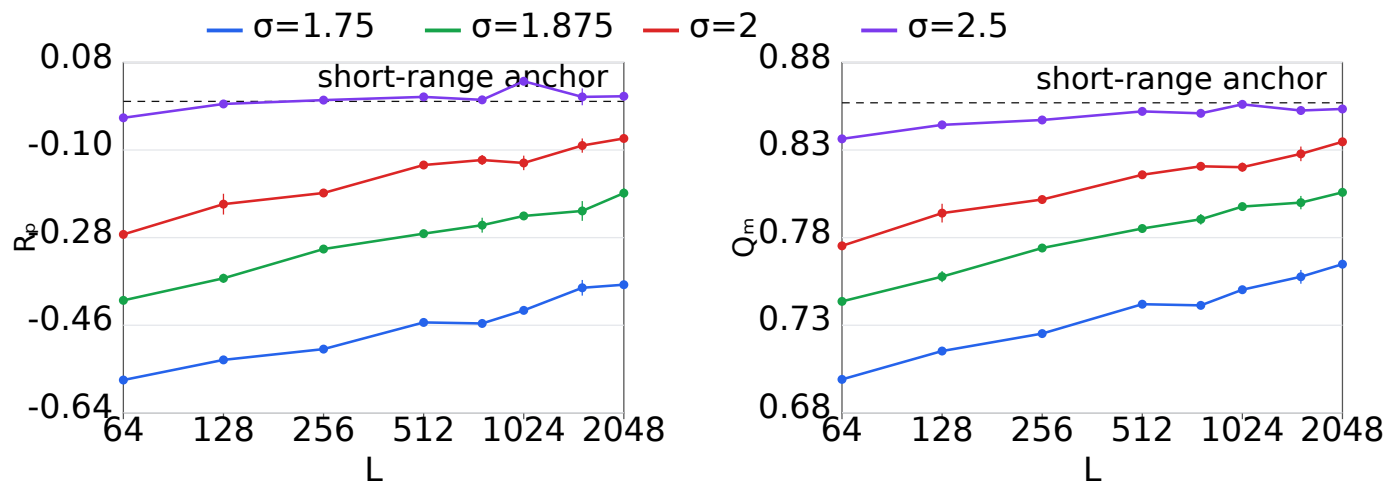
Physical meaning

- $\sigma = 1.75, \eta_{\text{eff}} = 0.3729$: largest deviation from the short-range benchmark.
- $\sigma = 1.875, \eta_{\text{eff}} = 0.3204$: substantial finite-size corrections.
- $\sigma = 2.0, \eta_{\text{eff}} = 0.2915$: closer to $\frac{1}{4}$, without locating the boundary.
- $\sigma = 2.5, \eta_{\text{eff}} = 0.2546$: consistent with the short-range control.

Extended $L \leq 2048$ evidence

Extending to $L=2048$ reveals a flow, not its endpoint

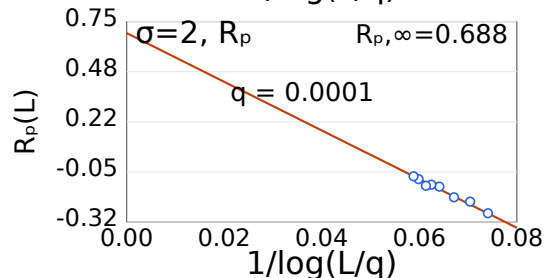
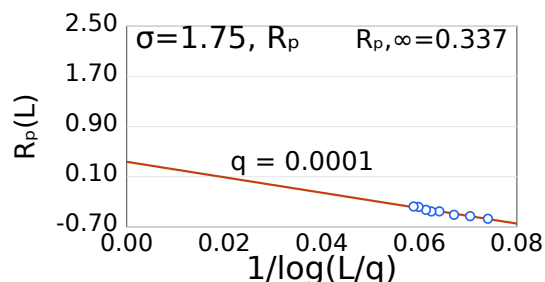
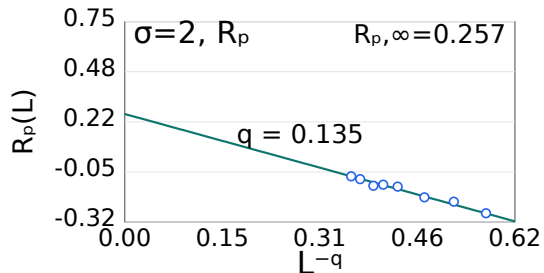
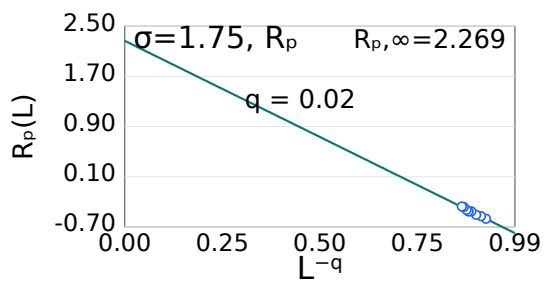
Figure 3 · Finite-size flow at the central critical points



Left: $R_p(L)$. Right: $Q_m(L)$. $\sigma = 1.75$ retains the clearest long-range signature; 1.875 and 2.0 drift slowly; 2.5 flows into the short-range Ising class.

The direction of flow is resolved; its asymptotic boundary is not.

Competing R_p extrapolations



$$AIC_c = -2 \ln \mathcal{L}_{\max} + 2k + \frac{2k(k+1)}{n-k-1}$$

n : data points

k : fitted parameters

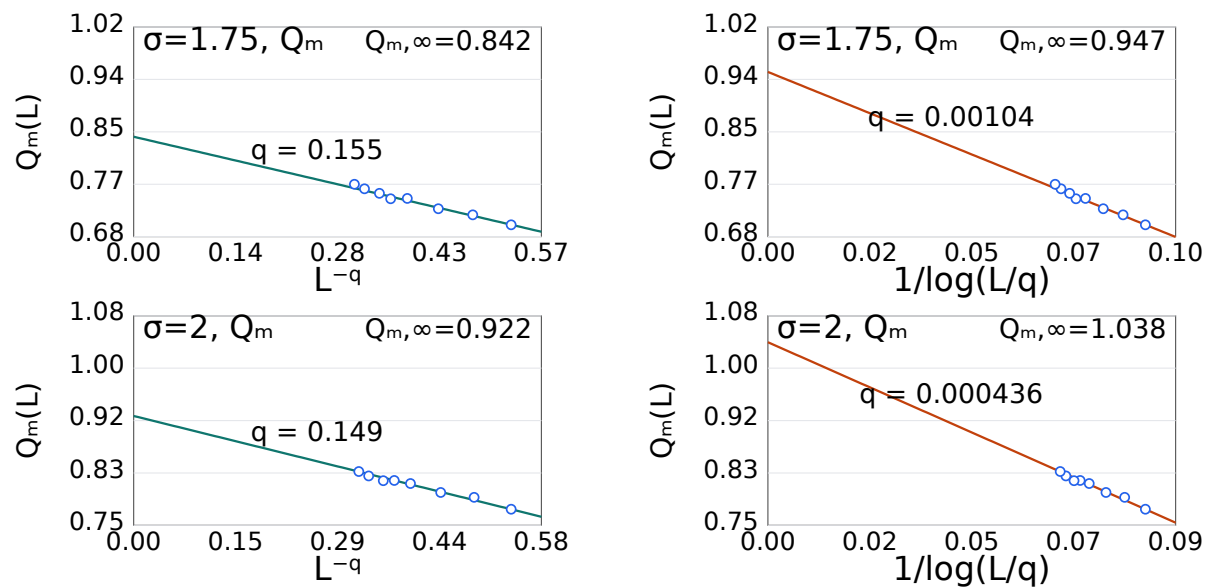
$\mathcal{L}_{\max}$: maximum likelihood

Lower AICc is preferred.

$\sigma = 1.75$: power has lower AICc, but $\frac{\chi^2}{dof} = 7.1$ and $R_{p,\infty} = 2.27$ make the limit noncredible.
 $\sigma = 2$: $|\Delta AICc| = 0.06$ leaves the endpoint ambiguous.

Competing Q_m extrapolations

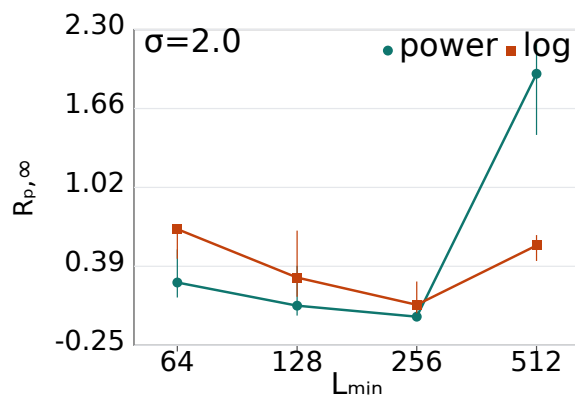
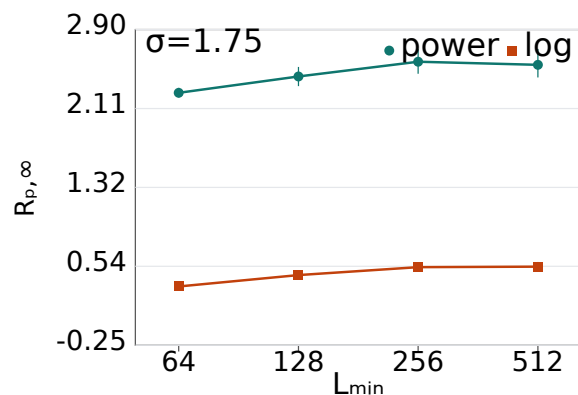
Figure 4b · Competing Q_m extrapolations



At the two candidate boundaries, $\sigma = 1.75$ and 2.0 , Q_m independently shows that neither visual linearity nor AICc selects a unique limit.

The R_p intercept is not window-stable

Figure 5a · R_p extrapolation-window stability



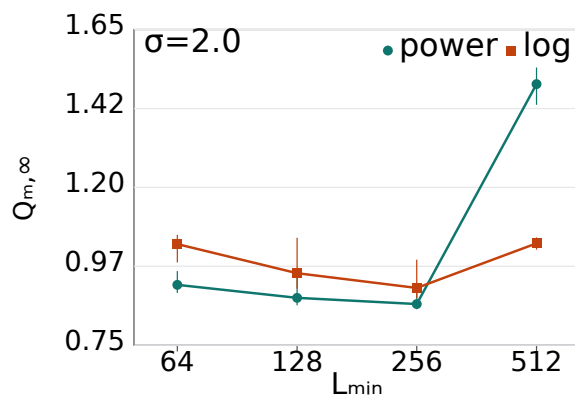
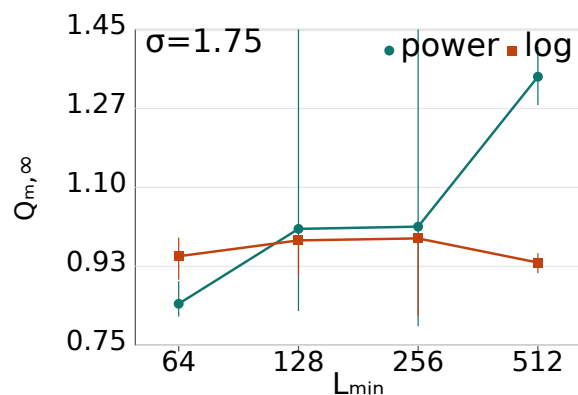
$$R_p = \langle R_2 \rangle - 2\langle R_0 \rangle$$

$$-2 \leq R_p \leq 1$$

A determined thermodynamic limit would remain stable as small sizes are removed. Instead, $R_{p,\infty}$ shifts strongly with the fit window, showing that the correction law—not the precision of individual Monte Carlo points—limits the inference.

The Q_m intercept is not window-stable

Figure 5b · Q_m extrapolation-window stability



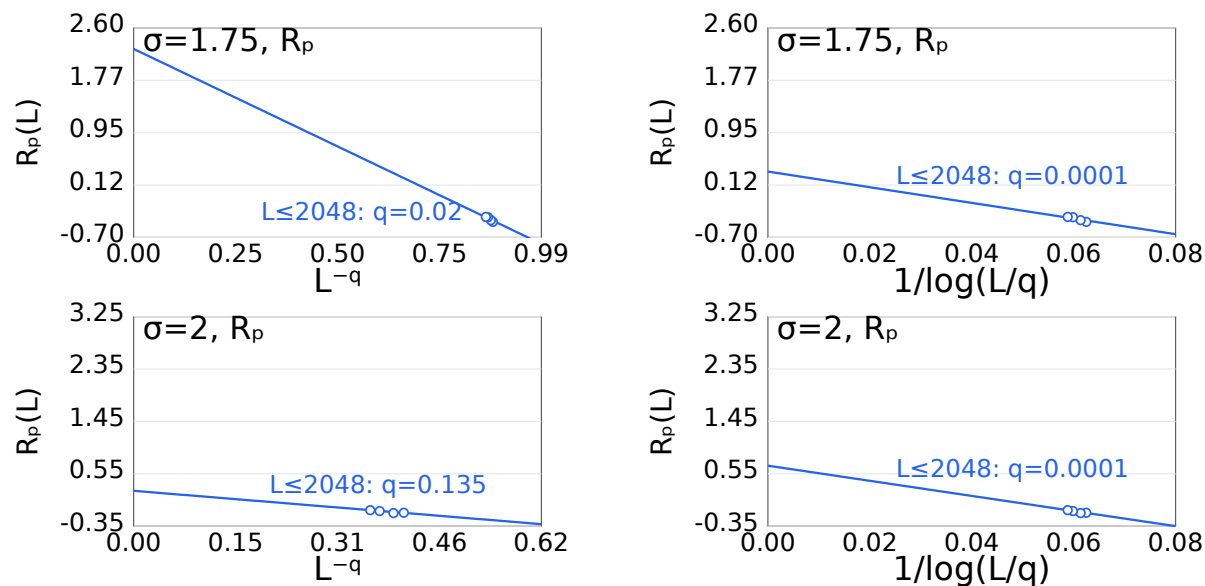
$$Q_m = \frac{\langle M^2 \rangle^2}{\langle M^4 \rangle}$$

$$0 \leq Q_m \leq 1$$

The measured $Q_{m(L)}$ values are precise, but they do not constrain a unique asymptotic limit. Removing small sizes makes the three-parameter extrapolation ill-conditioned, so the intercept drifts and its bootstrap interval can expand sharply.

Large-size R_p fits remain model-dependent

Figure 6a · Large-size R_p points and full-window fits

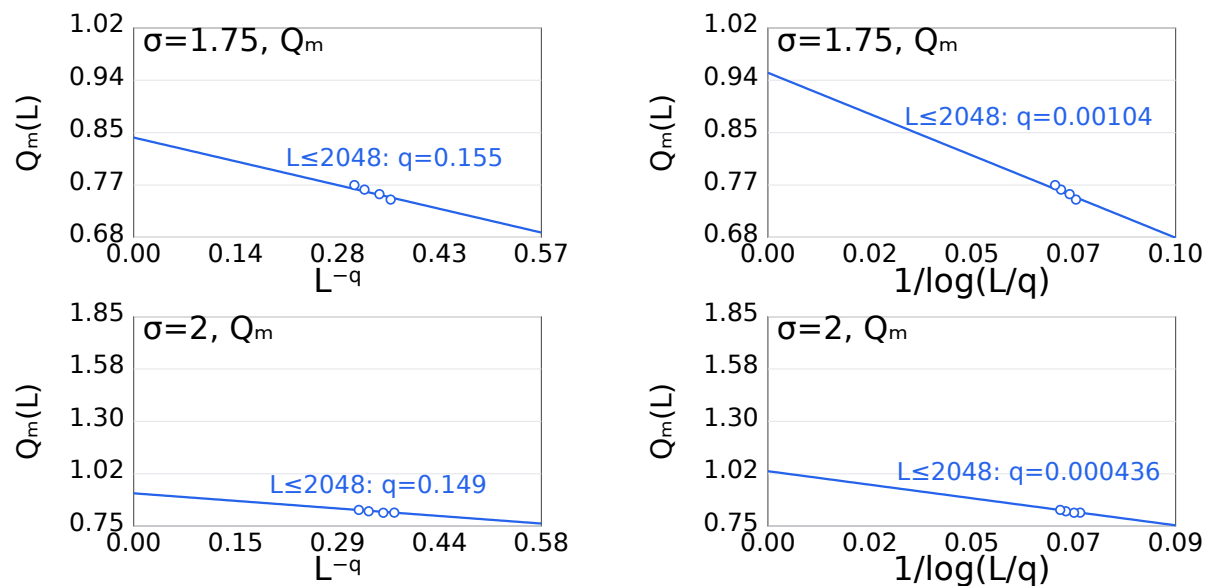


Only $L = 768, 1024, 1536, 2048$ points are shown; full-window fits move the pathological $\sigma = 2.0$ power intercept from $R_{p,\infty} = 3.06$ to 0.257 .

At $\sigma = 2$, larger sizes remove the worst power-fit pathology; $\sigma = 1.75$ remains ill-conditioned. At neither boundary do power and log agree.

Large-size Q_m fits remain model-dependent

Figure 6b · Large-size Q_m points and full-window fits



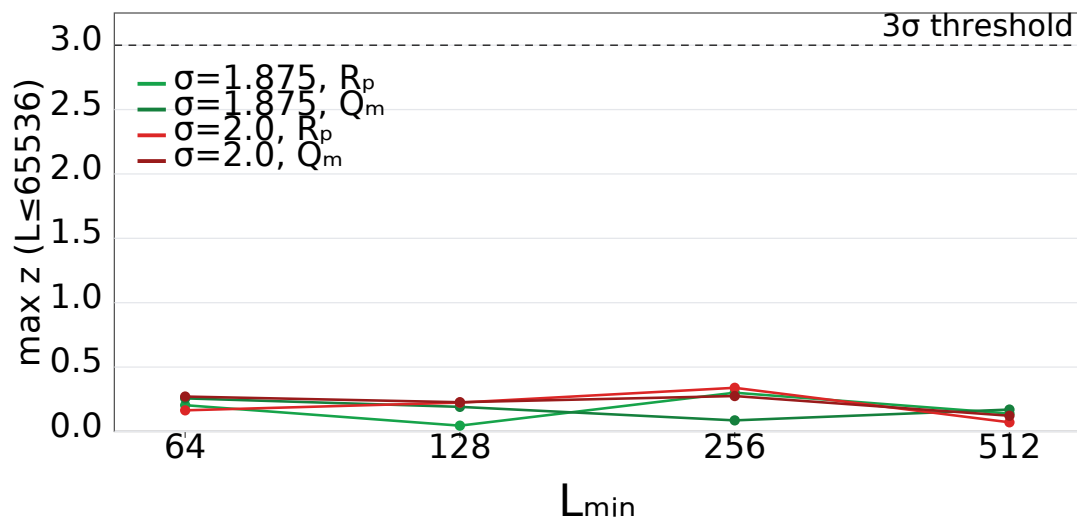
Only $L = 768, 1024, 1536, 2048$ points are shown; $Q_{m,\infty}$ moves from 1.76 to 0.922, but power and log limits still disagree.

At both candidate boundaries, the large-size Q_m points do not reconcile the power and logarithmic limits.

Discrimination and validation limits

A longer ladder alone still misses 3σ

Figure 7 · Distinguishable-size forecast



Every registered observable and fit window remains below 3σ through $L = 65536$.

Conditional error budget Included:

- parameter-prediction covariance;
- assumed future Monte Carlo standard error.

Not included:

- β_c systematics;
- additional model-form uncertainty.

>65536 is a planning result, not a claim about unmeasured physics.

The Clock cross-check identifies an algorithmic limit

σ	L	Qm difference	χ difference
1.875	64–256	$\leq 0.63\sigma$	$\leq 0.45\sigma$
2.0	64–256	$\leq 2.06\sigma$	$\leq 2.16\sigma$
1.875	512	-3.21σ	–
2.0	512	-4.38σ	–

Through $L = 256$, Clock and FK sample compatible equilibrium physics.

At $L = 512$, the discrepancy coincides with $\tau_{\text{int}} \approx 8 \times 10^3$, diagnosing local-update mixing failure.

An independent-algorithm comparison must audit autocorrelation, not only final means.

Conclusion

An evidence-backed inconclusive result

Validation

Pipeline is credible

The NN $\eta = \frac{1}{4}$ control passes; critical points and finite-size drift are reproduced.

Decision

Boundary remains open

Power/log and fit-window dependence prevent choosing $\sigma^* = \frac{7}{4}$ or 2.

Physics

The flow is resolved

$\sigma = 2.5$ reaches short-range anchors while $\sigma \leq 2$ retains strong corrections.

Next

Reduce corrections

Improve β_c and joint fits; seek weak-correction observables or an improved Hamiltonian that suppresses leading finite-size corrections.

The central result is not a selected boundary, but a quantitative account of why current scales cannot select one.